

Answer the following questions: (Marks)

Question 1: Answer the following (MCQ) questions: (Marks)

1- check bit errors.

A. nodal processing B. queueing delay C. transmission delay D. propagation delay

2- determine output link.

A. nodal processing B. queueing delay C. transmission delay D. propagation delay

3- time waiting at output link for transmission.

A. nodal processing B. queueing delay C. transmission delay D. propagation delay

4- depends on congestion level of router.

A. nodal processing B. queueing delay C. transmission delay D. propagation delay

5- $= L/R$, L : packet length (bits), R : link *bandwidth* (bps).

A. nodal processing B. queueing delay C. transmission delay D. propagation delay

6- $= d/s$, d : length of physical link, s : propagation speed in medium.

A. nodal processing B. queueing delay C. transmission delay D. propagation delay

7- self-replicating infection by receiving/executing object (e.g., e-mail attachment)

A. virus B. worm C. spyware D. botnet

8- self-replicating infection by passively receiving object that gets itself executed

A. virus B. worm C. spyware D. botnet

9- malware can record keystrokes, web sites visited, upload info to collection site.

A. virus B. worm C. spyware D. botnet

10- attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic.

A. virus B. worm C. spyware D. DoS

Question 2: Select which of the following statements are false and which are true?

- 1- RFC stands for Request for comments. () ✓
- 2- IETF stands for Internet Engineering Task Force. () ✓
- 3- protocols define format, order of msgs sent and received among network entities, and actions taken on msg transmission, receipt. () ✓
- 4- voice over DSL phone line goes to Internet. () ✗
- 5- data over DSL phone line goes to telephone net. () ✗
- 6- In DSL, voice, data transmitted at different frequencies over dedicated line to central office. () ✓
- 7- DSL stands for digital subscriber line. () ✓
- 8- In cable network, data, TV transmitted at different frequencies over shared cable distribution network. () ✓
- 9- link transmission rate, aka link *capacity*, aka link *bandwidth*. () ✓
- 10- packet transmission delay = time needed to transmit L -bit packet into link. () ✓
- 11- *twisted pair (TP)* is two insulated copper wires. () ✓
- 12- In packet-switching hosts break application-layer messages into *packets*. () ✓
- 13- If arrival rate (in bits) to link exceeds transmission rate of link for a period of time: packets will queue, wait to be transmitted on link. () ✓
- 14- If arrival rate (in bits) to link exceeds transmission rate of link for a period of time: packets can be dropped (lost) if memory (buffer) fills up. () ✓
- 15- *forwarding*: determines source-destination route taken by packets. () ✗
- 16- *routing*: move packets from router's input to appropriate router output. () ✗
- 17- connecting each access ISP to each other directly *does scale*. () ✗
- 18- R : link bandwidth (bps), L : packet length (bits), a : average packet arrival rate, $La/R \sim 0$: avg. queueing delay small. () ✓
- 19- R : link bandwidth (bps), L : packet length (bits), a : average packet arrival rate, $La/R \rightarrow 1$: avg. queueing delay large. () ✓
- 20- R : link bandwidth (bps), L : packet length (bits), a : average packet arrival rate, $La/R > 1$: more "work" arriving than can be serviced, average delay infinite! () ✓

21- lost packet may be retransmitted by previous node, by source end system, or not at all. () ✓

22-*throughput*: rate (bits/time unit) at which bits transferred between sender/receiver. () ✓

23-*instantaneous throughput*: rate at given point in time. () ✓

24-*average throughput*: rate over longer period of time. () ✓

25- In Internet protocol stack, *application*: supporting network applications

- FTP, SMTP, HTTP () ✓

26- In Internet protocol stack, *transport*: process-process data transfer

- TCP, UDP () ✓

27- In Internet protocol stack, *network*: routing of datagrams from source to destination

- IP, routing protocols () ✓

28- In Internet protocol stack, *link*: data transfer between neighboring network elements

- Ethernet, 802.111 (WiFi), PPP () ✓

29- In Internet protocol stack, *physical*: bits “on the wire”. () ✓

30- In ISO/OSI reference model, *presentation*: allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions. () ✓

31- In ISO/OSI reference model, *session*: synchronization, checkpointing, recovery of data exchange. () ✓

32-Internet stack missing the layers *presentation and session*. () ✓

33- Internet not originally designed with (much) security in mind. () ✗ ✓

34- infected host can be enrolled in botnet, used for spam. () ✓

35- *IP spoofing*: send packet with true source address. () ✗

36- server not always-on host. () ✗

37- server has permanent IP address. () ✓

38-server has data centers for scaling () ✓

39- In client-server architecture, clients communicate with server. () ✓

40- In client-server architecture, clients may be intermittently connected. () ✓

41- In client-server architecture, clients may have dynamic IP addresses. () ✓

42- In client-server architecture, clients do not communicate directly with each other.
() ✓

43- In P2P architecture, no always-on server. () ✓

44- In P2P architecture, arbitrary end systems directly communicate. () ✓

45- In P2P architecture, peers request service from other peers, provide service in return to other peers. () ✓

46- In P2P architecture, peers are intermittently connected and change IP addresses.
() ✓

47- HTTP is "stateless". () ✓

48- non-persistent HTTP: at most one object sent over TCP connection, connection then closed. () ✓

49- non-persistent HTTP: downloading multiple objects required multiple connections. ✓
()

50- persistent HTTP: multiple objects can be sent over single TCP connection between client, server. () ✓

51- RTT : time for a small packet to travel from client to server and back. () ✓

52- status code appears in 1st line in server-to-client response message. () ✓

53- status code: **200 OK**

- request succeeded, requested object later in this msg . () ✓

54- status code: **301 Moved Permanently**

- requested object moved, new location specified later in this msg
(Location:) () ✓

55- status code: **400 Bad Request**

- request msg not understood by server () ✓

56- status code: **404 Not Found**

- requested document not found on this server () ✓

57- status code: **505 HTTP Version Not Supported** () ✓